

Homework 08 One Way ANOVA SOLUTIONS

Due by 11:59pm, Friday, November 14, 2025

S&DS 2300/5300/ENV 757

The CSV file `HindiFilm.csv` which you can get [HERE](https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/HindiFilm.csv) (https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/HindiFilm.csv) contains the 1698 Hindi language movies that released in India across a 13 year period (2005-2017) (thanks to Premkumar, Prashant (2020), “Bollywood Movies data”, Mendeley Data, V1, doi: 10.17632/3c57btcxy9.1). Here are the variables:

- Name
- Period (normal or holiday)
- Remake (Yes/No)
- Franchise (Yes/No)
- Genre
- New_Actor (Yes/No)
- New_Director (Yes/No)
- New_Musician (Yes/No - i.e. new music director)
- Lead (Lead actor name)
- Director (Name)
- Musician (Music Director name)
- Screens (how many screens was it shown on)
- Revenue (Indian Rupees)
- Budget (Indian Rupees - usual shorthand is INR)

1) One Way ANOVA of percent return for movies. (75 pts - 6 pts each except part 1.12 which is 9 pts)

1.1) Read the data into an object called `movie` (do NOT use the option `as.is = TRUE`). Update this object so that it only contains the columns `Budget`, `Revenue`, `Genre`, and `Name`. In addition, retain only rows that have complete data for these four columns.

Next, create a new column called `pctReturn` that calculates the percentage return (have this be a percentage, NOT a fraction).

Make a table of `Genre` and sort this from high to low. Show these results.

Update `movie` so that it only contains movies in the following genres : “masala”, “drama”, “fantasy”, “love_story”, “action”.

As per usual, look at the first few rows of the data, get the dimension of the final dataset.

```
movie <- read.csv("https://raw.githubusercontent.com/jreuning/sds230_data/refs/heads/main/HindiFilm.csv")
movie <- movie[, c("Budget", "Revenue", "Genre", "Name")]
movie <- movie[complete.cases(movie), ]

movie$pctReturn <- movie$Revenue / movie$Budget * 100

sort(table(movie$Genre), decreasing = TRUE)
```

```
##
##      drama      comedy    thriller    love_story    action    rom__com
##      639       284       212       133       127       95
##      adult     horror     suspense     masala mythological     fantasy
##      78        53        30        16        14        13
##      animation documentary
##      3          1
```

```
keep_genres <- c("masala", "drama", "fantasy", "love_story", "action")
movie <- subset(movie, Genre %in% keep_genres)

head(movie)
```

```
##      Budget   Revenue  Genre      Name  pctReturn
## 2      825000 15000000  drama      Kaccha Limboo 1818.18182
## 4      4500000 210000000  drama      Qaidi Band 4666.66667
## 8      825000 15000000  drama      Future To Bright Hai Ji 1818.18182
## 9 1945820000 5200000000  action      Ghajini 26.72395
## 10 875785000 1800000000  drama      Taare Zameen Par 20.55299
## 11 525785000 3700000000  drama      Mangal Pandey – The Rising 70.37097
```

```
dim(movie)
```

```
## [1] 928 5
```

1.2) Make a boxplot of `pctReturn` by `Genre`. Superimpose a red dot at the mean percent return for each `Genre`; also print this mean rounded to the nearest percent. In addition, add a horizontal dotted blue line at 100 (this represents the place where revenue = budget). Discuss what you observe.

```
boxplot(pctReturn ~ Genre,
        data = movie,
        main = "Percent Return by Genre",
        ylab = "Percent Return",
        xlab = "Genre")

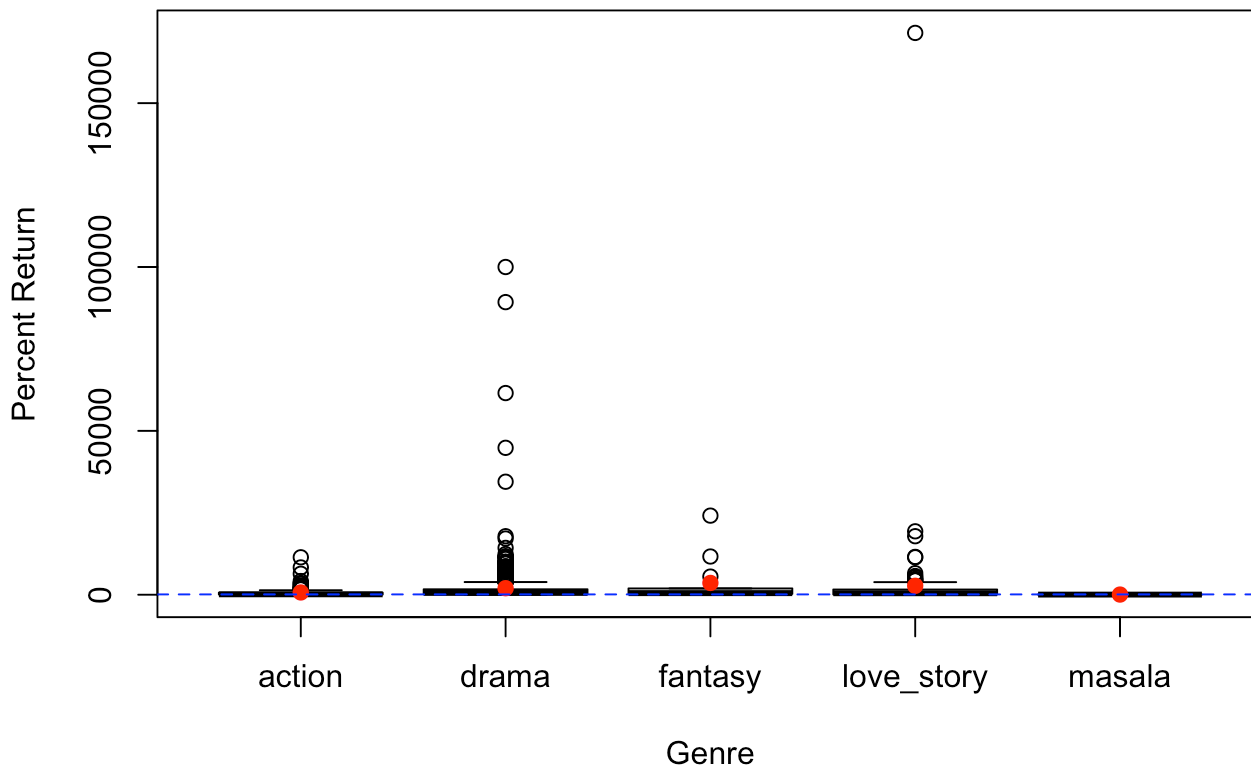
genre_means <- tapply(movie$pctReturn, movie$Genre, mean)
round(genre_means, 0)
```

```
##      action      drama      fantasy love_story      masala
##      692       1983       3601       2732       54
```

```
points(x = 1:length(genre_means),
       y = genre_means,
       col = "red",
       pch = 19)

abline(h = 100, col = "blue", lty = 2)
```

Percent Return by Genre



For each genre's mean percent return, it falls within a range of 54 and 3601. However, the data is not portrayed well in the boxplot, as the extreme outliers (going up to around 150000) make it so that the plots are too squished to distinguish sufficiently.

1.3) Make a new variable on 'movie' called `log10pct` that is log base 10 of percent return (the function you want is `log10()`). Repeat part 1.2, but modify the horizontal line so that it is at the correct value (when rounding mean `log10(pct return)`, round to 1 decimal place). Discuss what you observe.

```
movie$log10pct <- log10(movie$pctReturn)
genre_means_log <- tapply(movie$log10pct, movie$Genre, mean)
round(genre_means_log, 1)
```

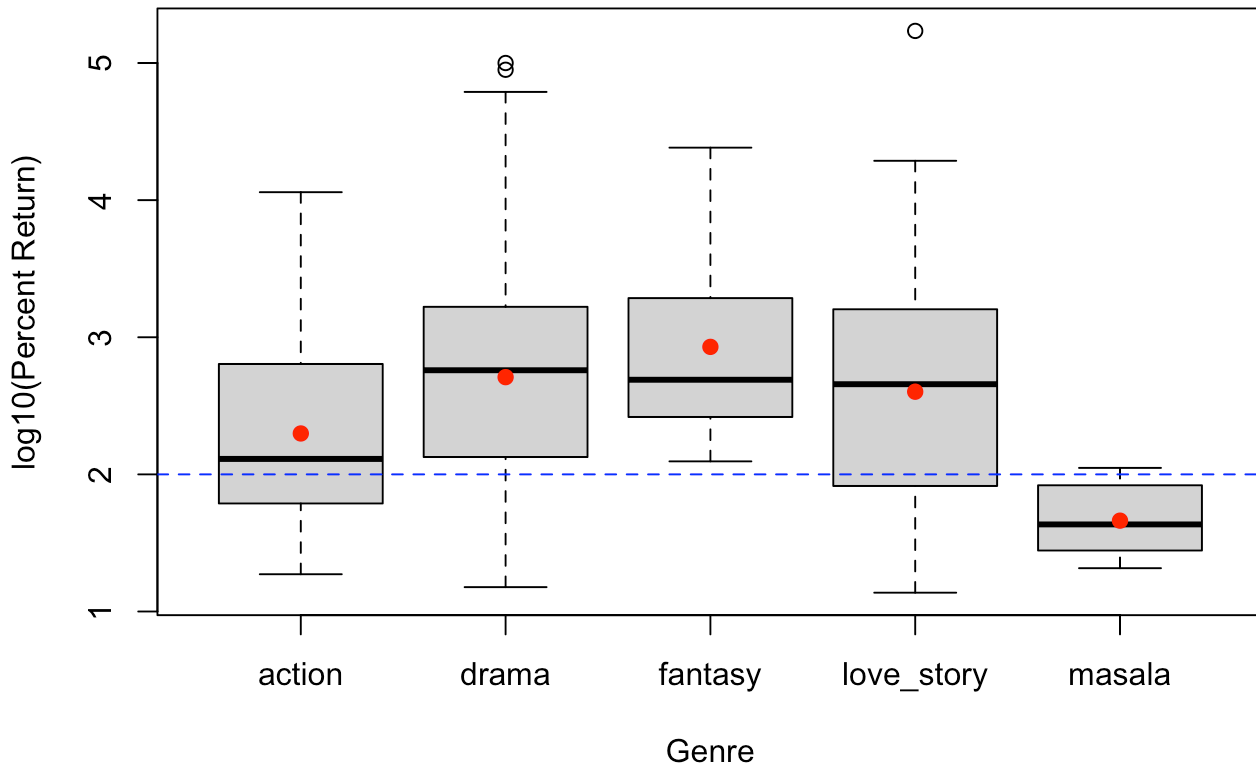
##	action	drama	fantasy	love_story	masala
##	2.3	2.7	2.9	2.6	1.7

```
boxplot(log10pct ~ Genre,
        data = movie,
        main = "log10(Percent Return) by Genre",
        ylab = "log10(Percent Return)",
        xlab = "Genre")
```

```
points(x = 1:length(genre_means_log),
       y = genre_means_log,
       col = "red",
       pch = 19)
```

```
abline(h = log10(100), col = "blue", lty = 2)
```

log10(Percent Return) by Genre

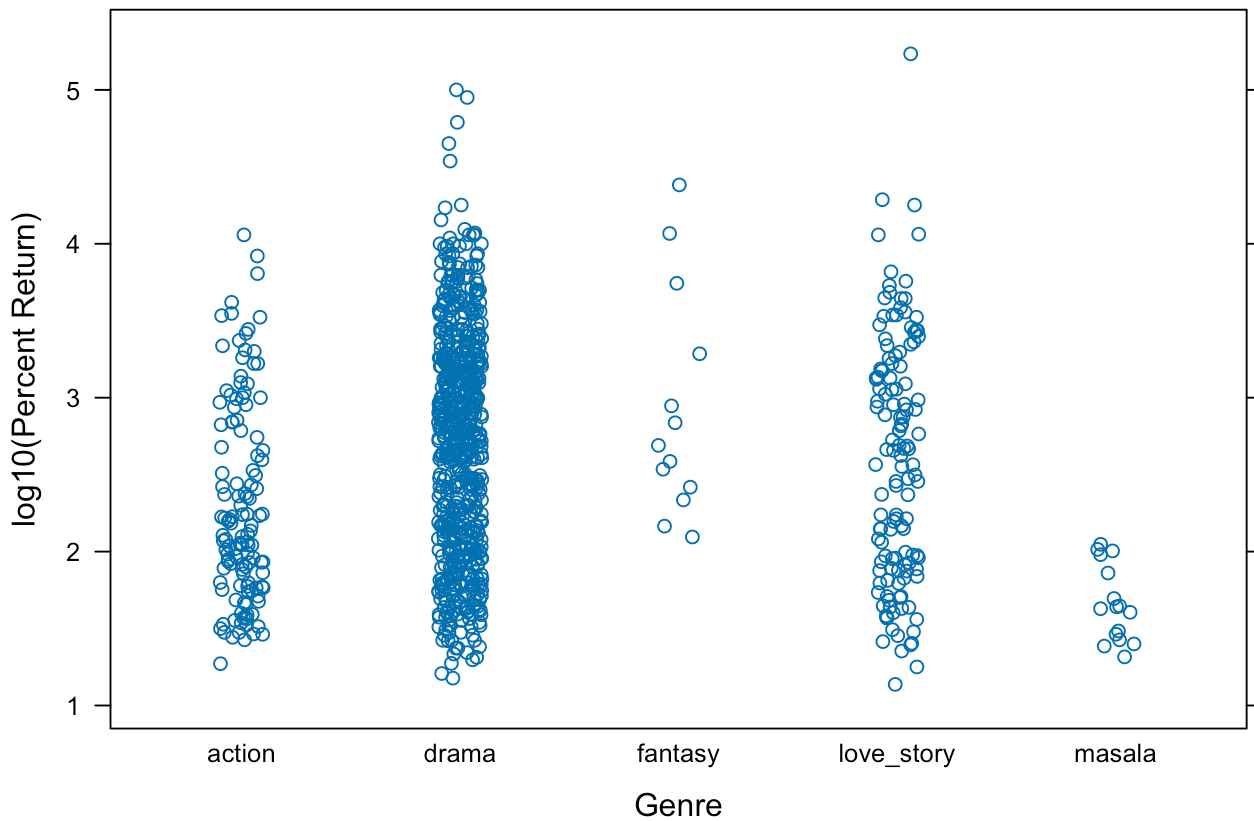


Here, the boxplot is not squished. Generally, we see that the mean and median for most genres at least break even. The only genre whose mean and median is below 100% pctReturn is masala.

1.4) Make a stripplot of log10pct by Genre. Which plot do you find more instructive - the boxplot or the strip plot (no right answer here, but give a one sentence reason for your opinion)?

```
library(lattice)
stripplot(log10pct ~ Genre,
  data = movie,
  jitter = 0.1,
  main = "log10(Percent Return) by Genre",
  ylab = "log10(Percent Return)",
  xlab = "Genre")
```

log10(Percent Return) by Genre



I think the boxplot is more instructive because while the stripplot better displays the count, the boxplot better displays measurements such as median, IQR, outliers, etc.

1.5) Just using the visual information provided by the boxplot of log10pct, do you think that the distribution of observations inside each Genre is approximately normal? Do you think the standard deviation is approximately the same in each Genre? (Remember that ANOVA assumes we have a normal distribution inside each Genre and that the standard deviation is the same across groups.)

Based on the boxplot, I think that the distribution of observations inside each genre is normal mostly based on where the IQR is. However, I do not think the standard deviation is approximately the same in each genre because masala is a lot more squished than other genres.

1.6) Calculate the sample standard deviation of log10pct for each Genre. Calculate the ratio of largest to smallest sample standard deviation. Is it reasonable to assume that the variances are the same across Genres?

```
sd_by_genre <- tapply(movie$log10pct, movie$Genre, sd)
sd_by_genre
```

```
##      action      drama    fantasy love_story      masala
## 0.6441561 0.7210816 0.7319048 0.7944657 0.2487290
```

```
sd_ratio <- max(sd_by_genre) / min(sd_by_genre)
sd_ratio
```

```
## [1] 3.194101
```

It is unreasonable to assume that the variances are the same across genres, since the largest log10pct standard deviation is 3.19 times as large as the smallest, which is slightly above the typical threshold of 2.

1.7) Use the `aov()` function to compare mean `log10pct` between Genres. Save your results to an object called `aov1`. Get summary information for `aov1`. Is the mean `log10pct` return statistically significantly different between Genres? In addition, confirm that the degrees of freedom reported by the test are what you expect (and write a sentence about this).

```
aov1 <- aov(log10pct ~ Genre, data = movie)
summary(aov1)
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Genre          4   34.2    8.539    16.6 3.79e-13 ***
## Residuals     923  474.7    0.514
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The mean `log10pct` is statistically significantly different between genres because the $Pr(>F)$ is $3.79e-13$, much less than the standard 0.05 threshold. For genre and residuals, $df = 4$ and $df = 923$, respectively. This makes sense because we have 5 genres ($5 - 1 = 4$) and 928 residuals ($928 - 5 = 923$).

1.8) Fit `log10pct` return based on Genre as a regression model WITHOUT an intercept. Save the results to an object called `mod1`. Following the example code provided in class, calculate confidence intervals for the mean percent return for each Genre and report results to two decimal places. Then make a plot of the resulting confidence intervals using the `plotCI()` function in the `plotrix` library.

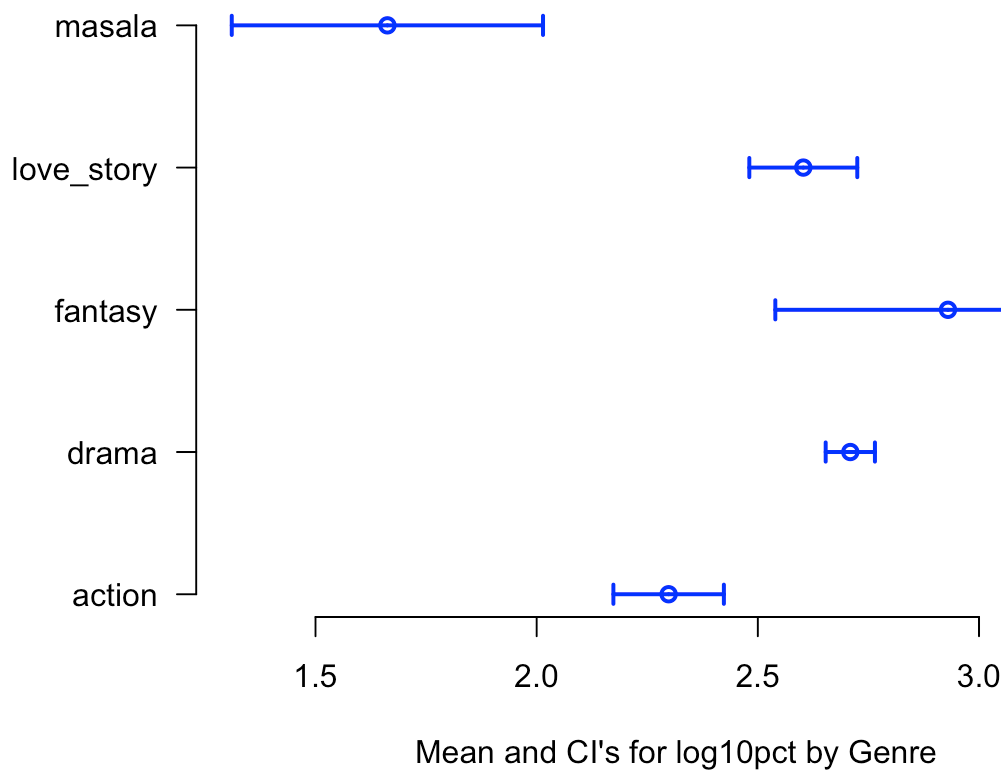
You'll notice that some intervals overlap while others do not. What should you conclude from this?

```
mod1 <- lm(log10pct ~ Genre - 1, data = movie)
CIs <- confint(mod1)
round(CIs, 2)
```

```
##              2.5 % 97.5 %
## Genreaction    2.17   2.42
## Genredrama     2.65   2.76
## Genrefantasy   2.54   3.32
## Genrelove_story 2.48   2.72
## Genremasala    1.31   2.01
```

```
coefs <- coef(mod1)

library(plotrix)
par(mar=c(5,8,4,2))
plotCI(coefs, 1:(length(coefs)), ui = CIs[,2], li = CIs[,1], axes = FALSE, err = "x", ylab = "",
       xlab = "Mean and CI's for log10pct by Genre", lwd = 2, col = "blue")
axis(side = 1)
axis(side = 2, at = 1:length(coefs), label = sub("Genre", "", names(coefs)), las = 2)
```



Nothing should be concluded based on which intervals overlap and which do not.

1.9) Use the `pairwise.t.test()` function to calculate Holm's correction for comparing pairs of means. Which pairs of Genres have statistically significantly different percent returns (use $\alpha = 0.05$)?

```
pairwise.t.test(movie$log10pct, movie$Genre)
```

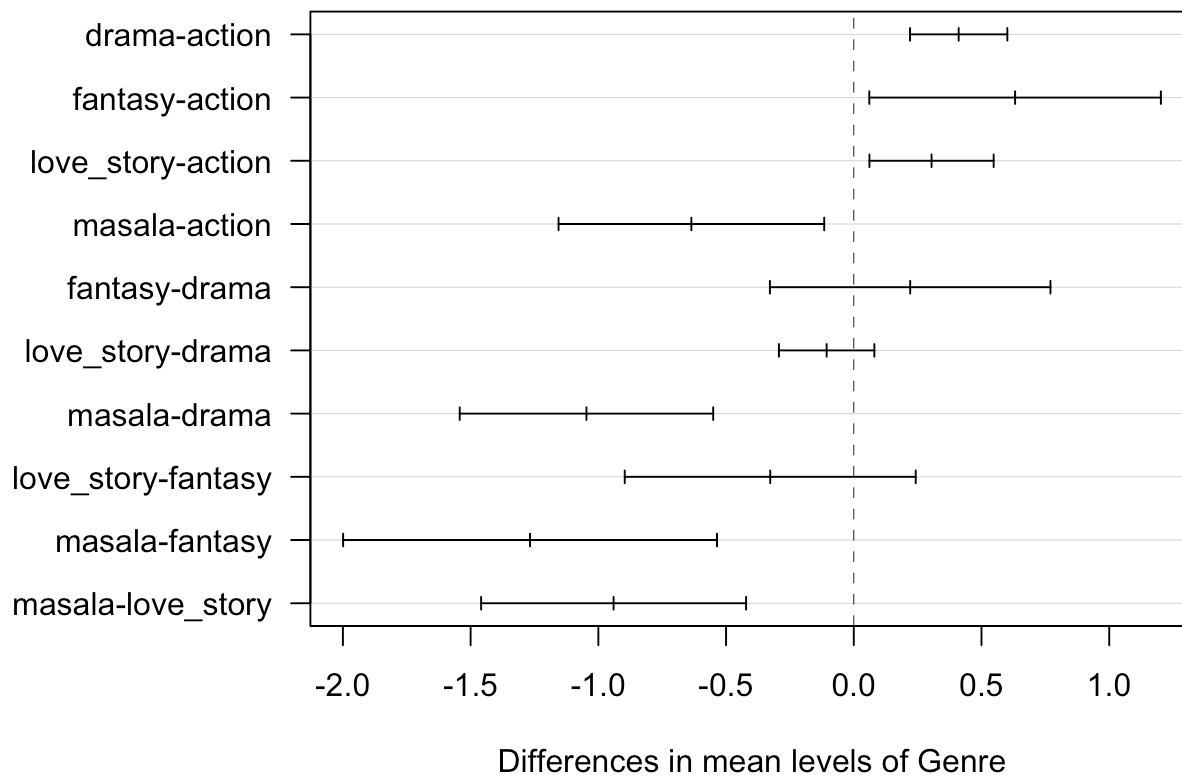
```
##
## Pairwise comparisons using t tests with pooled SD
##
## data: movie$log10pct and movie$Genre
##
##          action  drama  fantasy love_story
## drama    5.3e-08 -      -          -
## fantasy   0.0103 0.3506 -          -
## love_story 0.0039 0.3506 0.3506 -
## masala    0.0043 1.0e-07 1.8e-05 6.9e-06
##
## P value adjustment method: holm
```

Drama + action, fantasy + action love_story + action, masala + action, masala + drama, masala + fantasy, and masala + love_story all have statistically significantly different percent returns.

1.10) Calculate Tukey simultaneous 95% confidence intervals for differences in mean percent return using the `TukeyHSD()` function. Plot the resulting confidence intervals. Do you reach the same conclusions as when using the Holm correction?

```
par(mar = c(5,11,4,1))
plot(TukeyHSD(aov1), las = 1)
```

95% family-wise confidence level



We get the same conclusions as when using the Holm correction. Fantasy+drama, love_story+drama, and love_story+fantasy all have CIs that overlap with 0, meaning that they are not statistically significantly different, matching the results of the Holm correction.

1.11) Create residual plots appropriate for `aov1`. Does it appear that the residuals have an approximately normal distribution? Is this good or bad? Comment in a sentence or two about the plot of fits versus residuals.

```
library(car)
```

```
## Loading required package: carData
```

```

myResPlots <- function(model, label){
  #plot of fitted vs. studentized residuals
  plot(rstudent(model) ~ model$fitted.values, pch = 19, col = 'red',
       xlab = "Fitted Values", ylab = "Studentized Residuals",
       main = paste("Fits vs. Studentized Residuals,", label))
  abline(h = 0, lwd = 3)
  abline(h = c(2,-2), lty = 2, lwd = 2, col="blue")
  abline(h = c(3,-3), lty = 2, lwd = 2, col="green")

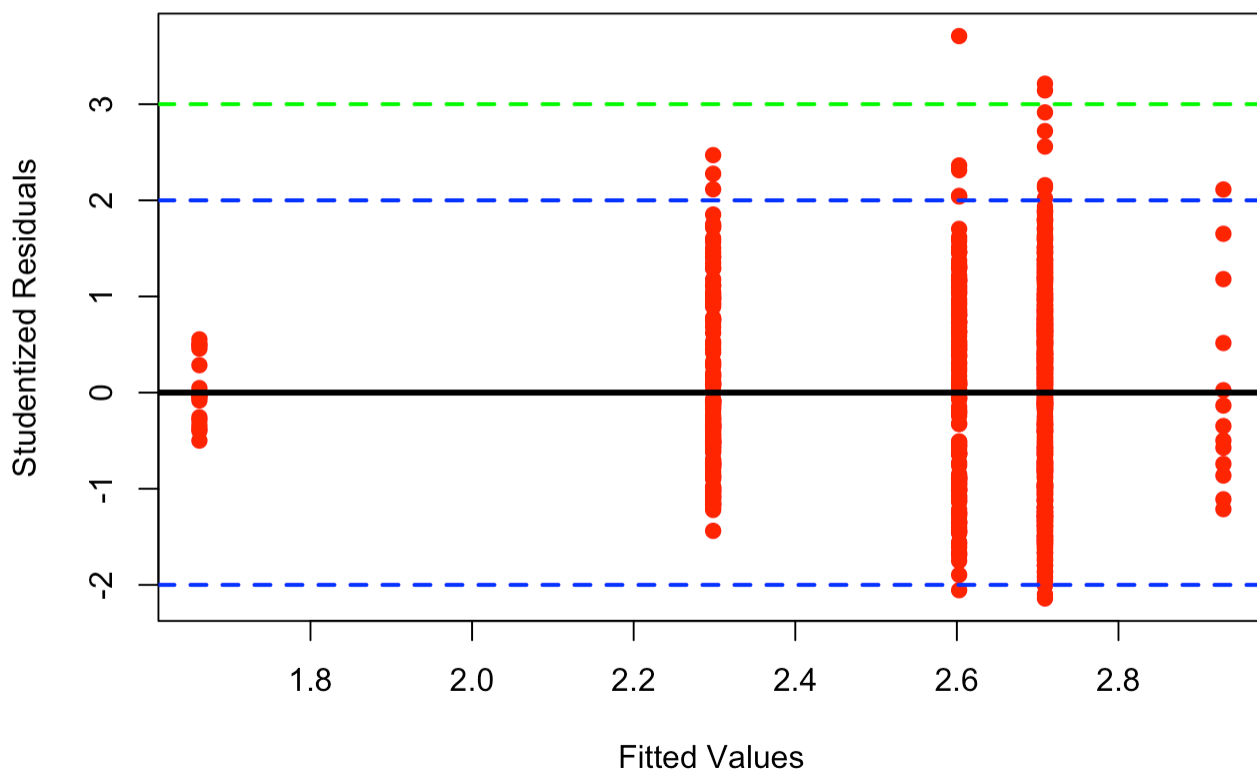
  #Normal quantile plot of studentized residuals
  qqplot(rstudent(model), pch = 19, main = paste("NQ Plot of Studentized Residuals,", label))

  #Plot of Cooks Distances
  plot(model, which = 4, lwd = 3, col = 'red')
  abline(h = 6/length(model$residuals), col = 'black', lwd = 3, lty = 2)
}

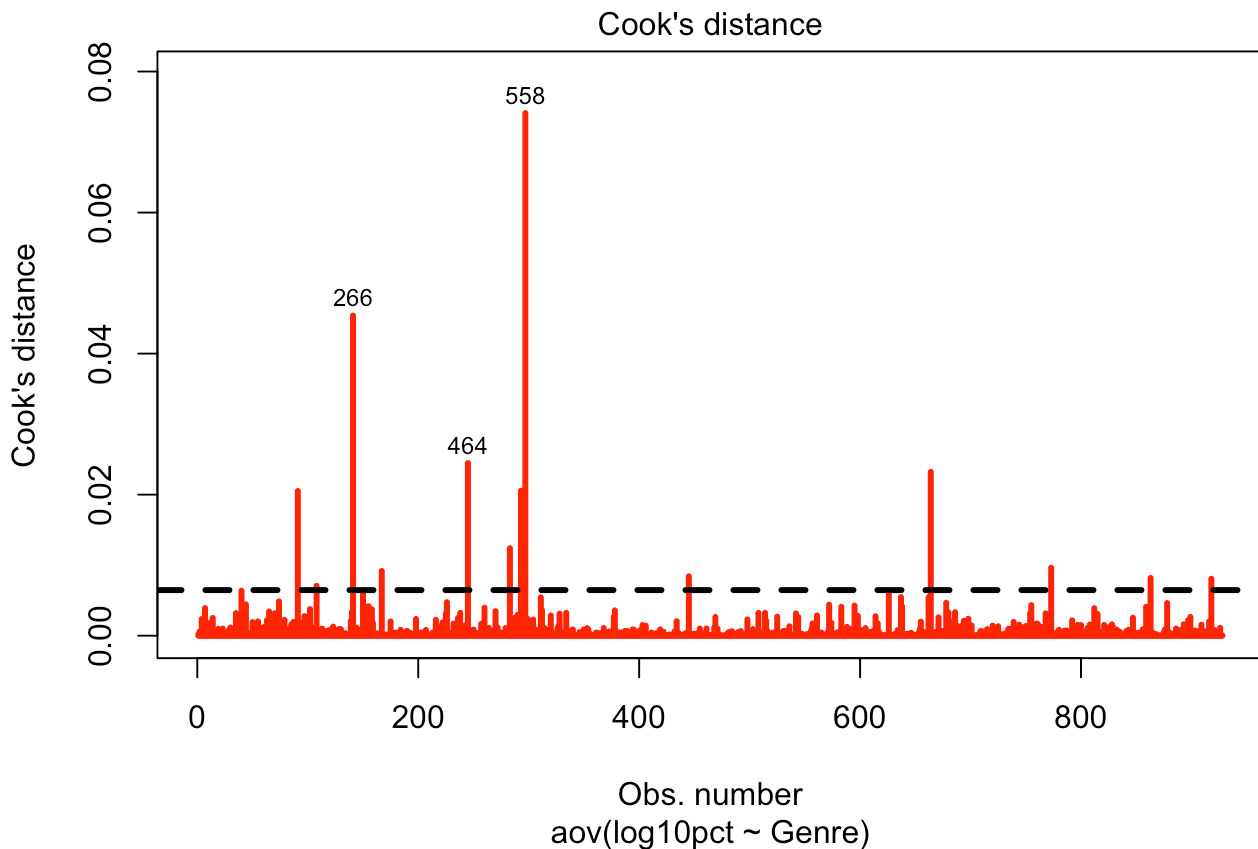
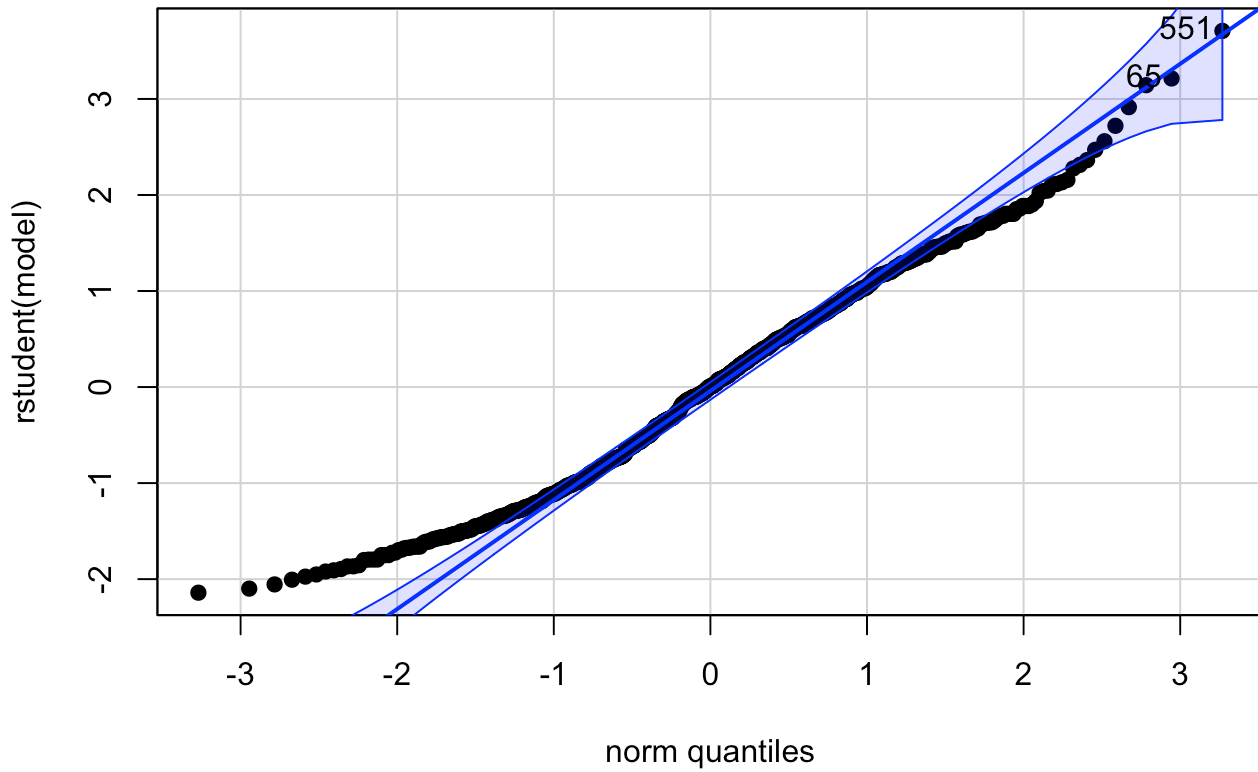
myResPlots(aov1, label = "log10pct")

```

Fits vs. Studentized Residuals, log10pct



NQ Plot of Studentized Residuals, log10pct



I'd say that the residuals are somewhat close to a normal distribution (which is good), with the main concerns being the left tail end going outside of the expected range (makes it not as good). As for fits vs residuals, we can see that the variance is not constant as there's a funnel shape.

1.12) Just to see what would have happened if we didn't use a log10 transformation, repeat parts 1.7 through 1.11 on the original percent return data (i.e. not log10). Just use one block of code (literally, just copy your code, replace `log10pct` with `pctReturn`).

Write about three sentences commenting on the fit of this model vs. the fit of the previous model. Specifically, comment on whether any pairs of groups changed significance, the overall model significance, and the residual plots.

```
aov2 <- aov(pctReturn ~ Genre, data = movie)
summary(aov2)
```

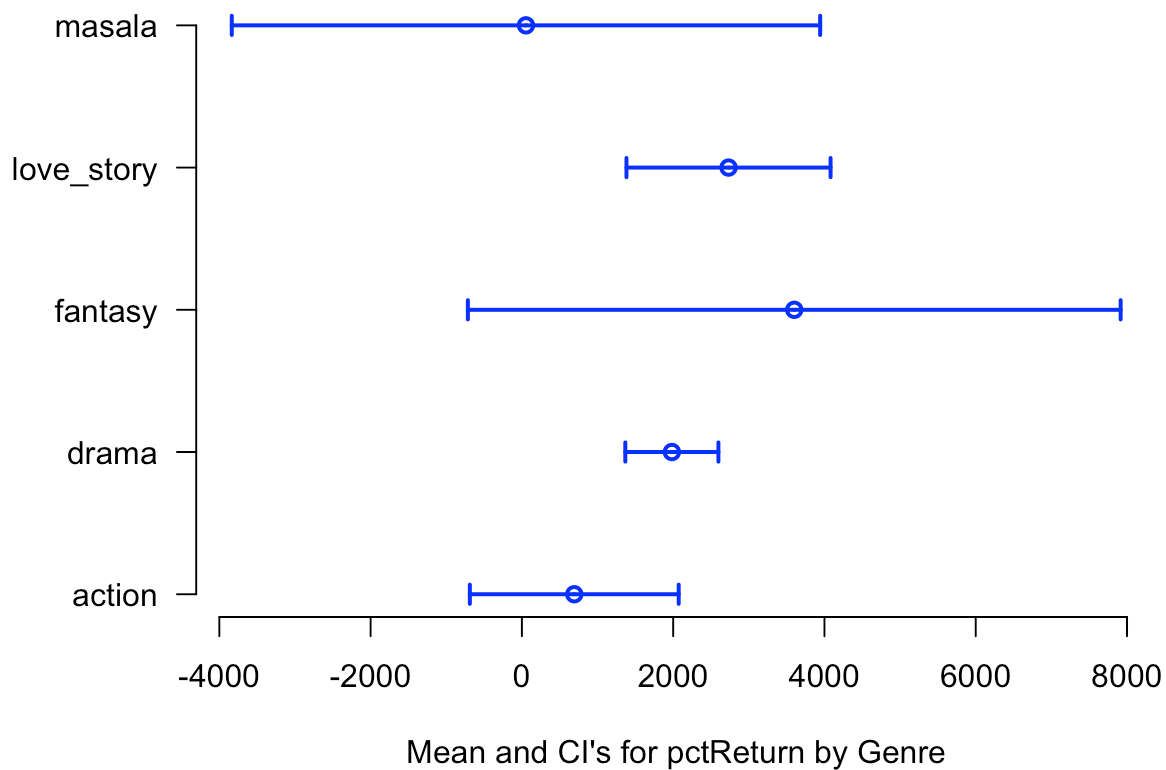
```
##              Df    Sum Sq Mean Sq F value Pr(>F)
## Genre         4 3.737e+08 93431520   1.487  0.204
## Residuals    923 5.801e+10 62847159
```

```
mod2 <- lm(pctReturn ~ Genre - 1, data = movie)
CIs <- confint(mod2)
round(CIs, 2)
```

```
##              2.5 %  97.5 %
## Genreaction   -688.17 2072.97
## Genredrama     1367.27 2598.22
## Genrefantasy   -714.30 7915.87
## Genrelove_story 1382.71 4080.85
## Genremasala    -3835.75 3943.37
```

```
coefs <- coef(mod2)

library(plotrix)
par(mar=c(5,8,4,2))
plotCI(coefs, 1:(length(coefs)), ui = CIs[,2], li = CIs[,1], axes = FALSE, err = "x", ylab = "",
       xlab = "Mean and CI's for pctReturn by Genre", lwd = 2, col = "blue")
axis(side = 1)
axis(side = 2, at = 1:length(coefs), label = sub("Genre", "", names(coefs)), las = 2)
```

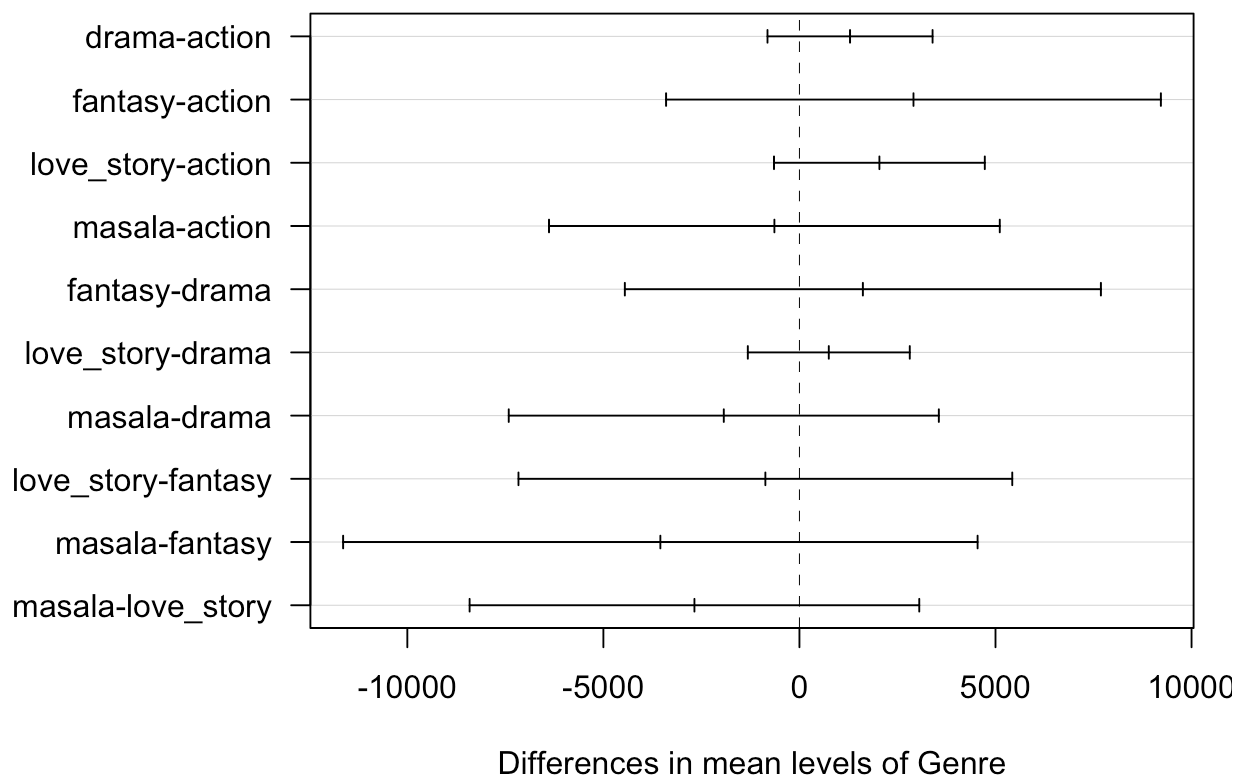


```
pairwise.t.test(movie$pctReturn, movie$Genre)
```

```
##
## Pairwise comparisons using t tests with pooled SD
##
## data: movie$pctReturn and movie$Genre
##
##      action drama fantasy love_story
## drama  0.85  -      -      -
## fantasy 1.00  1.00  -      -
## love_story 0.38  1.00  1.00  -
## masala  1.00  1.00  1.00  1.00
##
## P value adjustment method: holm
```

```
par(mar = c(5,11,4,1))
plot(TukeyHSD(aov2), las = 1)
```

95% family-wise confidence level



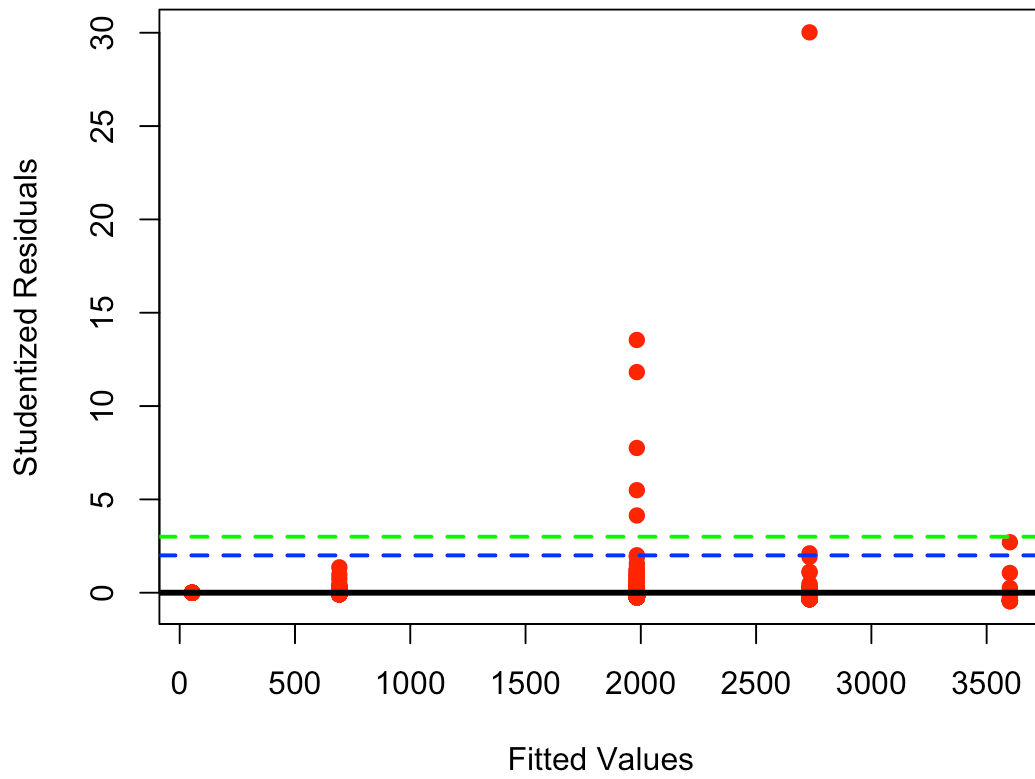
```
library(car)
myResPlots <- function(model, label){
  #plot of fitted vs. studentized residuals
  plot(rstudent(model) ~ model$fitted.values, pch = 19, col = 'red',
       xlab = "Fitted Values", ylab = "Studentized Residuals",
       main = paste("Fits vs. Studentized Residuals,", label))
  abline(h = 0, lwd = 3)
  abline(h = c(2,-2), lty = 2, lwd = 2, col="blue")
  abline(h = c(3,-3), lty = 2, lwd = 2, col="green")

  #Normal quantile plot of studentized residuals
  qqplot(rstudent(model), pch = 19, main = paste("NQ Plot of Studentized Residuals,", label))

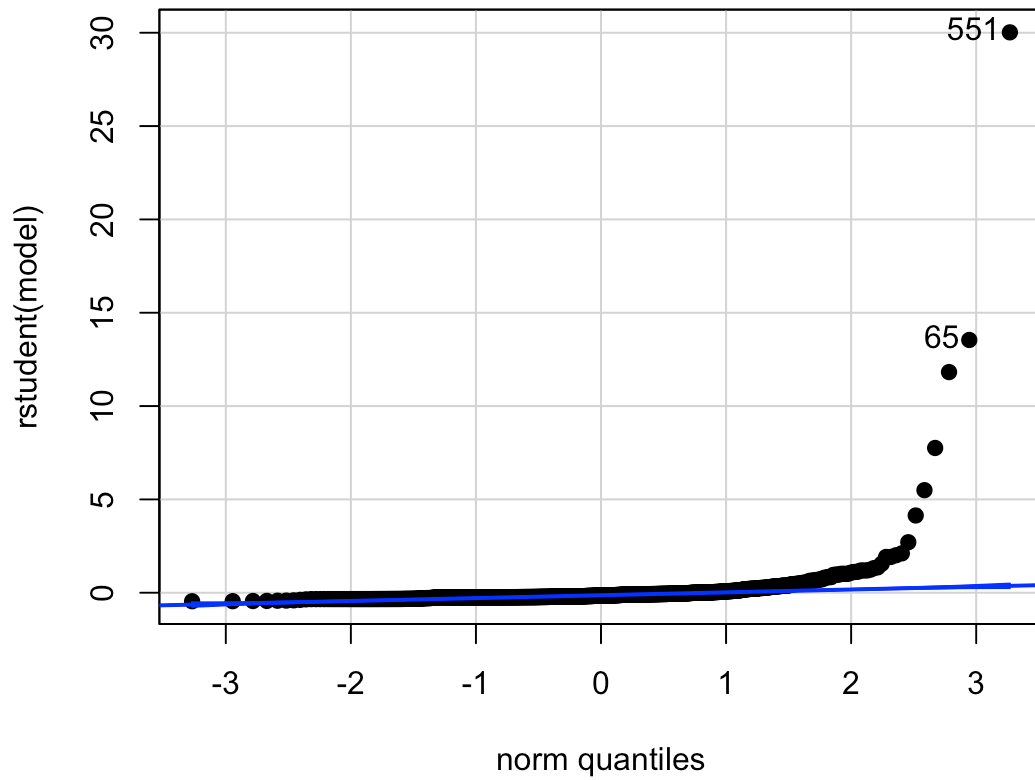
  #Plot of Cooks Distances
  plot(model, which = 4, lwd = 3, col = 'red')
  abline(h = 6/length(model$residuals), col = 'black', lwd = 3, lty = 2)
}

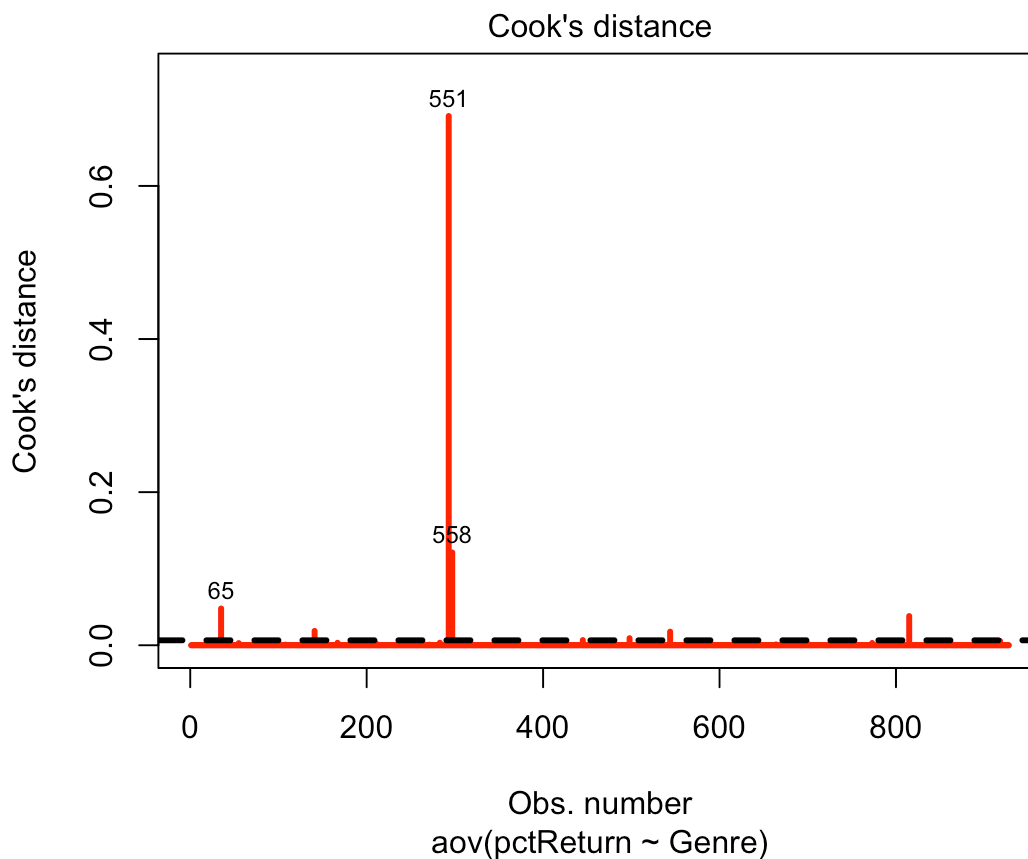
myResPlots(aov2, label = "pctReturn")
```

Fits vs. Studentized Residuals, pctReturn



NQ Plot of Studentized Residuals, pctReturn





The non-log transformed model fits quite poorly. When looking at the p values between groups, none of them reached below the 0.05 significance threshold, and genre had a $Pr(>F) = 0.204$, much larger than what it was when we fit $\log_{10}pct$. Additionally, both fits vs residuals and qq plot showed assumption violations (heteroscedasticity and not normal).

2) Non-parametric tests. (25 pts - 5pts each)

2.1) Perform a Bartlett test to see whether variances of percent return are the same across Genres. Then, repeat for $\log_{10}pct$. Write one sentence about what you conclude.

```
bartlett.test(movie$pctReturn, movie$Genre)
```

```
##
## Bartlett test of homogeneity of variances
##
## data:  movie$pctReturn and movie$Genre
## Bartlett's K-squared = 649.27, df = 4, p-value < 2.2e-16
```

```
bartlett.test(movie$log10pct, movie$Genre)
```

```
##
## Bartlett test of homogeneity of variances
##
## data:  movie$log10pct and movie$Genre
## Bartlett's K-squared = 23.962, df = 4, p-value = 8.129e-05
```

Both Bartlett tests have a $p_value < 0.05$, meaning that variances are significantly different (though doing a log transformation helped reduce this difference).

2.2) Perform a Levene test to see whether variances of percent return are the same across Genres. Then, repeat for `transpct`. Write one sentence about what you observe (be sure to review the characteristics of Levene's test).

```
leveneTest(movie$pctReturn, movie$Genre)
```

```
## Warning in leveneTest.default(movie$pctReturn, movie$Genre): movie$Genre
## coerced to factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value Pr(>F)
## group  4  1.3838 0.2376
##      923
```

```
leveneTest(movie$log10pct, movie$Genre)
```

```
## Warning in leveneTest.default(movie$log10pct, movie$Genre): movie$Genre coerced
## to factor.
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##      Df F value    Pr(>F)
## group  4  6.4329 4.167e-05 ***
##      923
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Levene Test shows that variance IS NOT significantly different when using `pctReturn`, but IS significantly different when using `log10pct`, and this result is due to the fact that the Levene test does not assume the observations are from normal distributions, a condition that was assumed for Bartlett tests.

2.3) Perform Welch's ANOVA on the original data which you will recall assumes unequal variances but which assumes normal distributions in each group (i.e. compare `pctReturn` by `Genre`). Compare results to regular one-way ANOVA. Is Welch's ANOVA a good choice here?

```
oneway.test(pctReturn ~ Genre, data = movie)
```

```
##
## One-way analysis of means (not assuming equal variances)
##
## data:  pctReturn and Genre
## F = 20.711, num df = 4.000, denom df = 79.718, p-value = 9.859e-12
```

```
summary.aov(aov(pctReturn ~ Genre, data = movie))
```

```
##      Df    Sum Sq Mean Sq F value Pr(>F)
## Genre    4 3.737e+08  93431520   1.487  0.204
## Residuals 923 5.801e+10 62847159
```

Welch's ANOVA detects significant differences in mean `pctReturn` between genres ($p\text{-value} = 9.859e-12$) while regular one-way ANOVA does not ($Pr(>F) = 0.204$). However, we know that the distribution of our data on the raw scale is not normally distributed in each group which is an assumption of Welch's ANOVA. Thus, regardless of the results, neither regular ANOVA or

Welch's ANOVA is a good choice on the raw scale.

2.4) Perform a non-parametric Kruskal Wallis test which recall makes NO assumptions about equal variances or about normality. Compare results to regular one-way ANOVA. Is Kruskal Wallis a good choice here?

```
kruskal.test(pctReturn ~ Genre, data = movie)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  pctReturn by Genre
## Kruskal-Wallis chi-squared = 63.473, df = 4, p-value = 5.397e-13
```

```
summary.aov(aov(pctReturn ~ Genre, data = movie))
```

```
##              Df    Sum Sq Mean Sq F value Pr(>F)
## Genre         4 3.737e+08 93431520   1.487  0.204
## Residuals    923 5.801e+10 62847159
```

Kruskal Wallis Test detects significant differences in mean pctReturn between genres (p-value = 9.859e-12) while regular one-way ANOVA does not ($Pr(>F) = 0.204$). Kruskal Wallis is a good choice here because pctReturn does not follow normality nor equal variances, assumptions used in regular one-way ANOVA but not in Kruskal Wallis.

2.5) Repeat 2.3) and 2.4) on the log10 scale. Discuss your results and compare results to regular ANOVA.

```
oneway.test(log10pct ~ Genre, data = movie)
```

```
##
##  One-way analysis of means (not assuming equal variances)
##
## data:  log10pct and Genre
## F = 61.337, num df = 4.000, denom df = 55.497, p-value < 2.2e-16
```

```
kruskal.test(log10pct ~ Genre, data = movie)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  log10pct by Genre
## Kruskal-Wallis chi-squared = 63.473, df = 4, p-value = 5.397e-13
```

```
summary.aov(aov(log10pct ~ Genre, data = movie))
```

```
##              Df Sum Sq Mean Sq F value    Pr(>F)
## Genre         4   34.2    8.539    16.6 3.79e-13 ***
## Residuals    923  474.7     0.514
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

All three tests are significant, with p values going from lowest to highest being Welch's ANOVA < one-way ANOVA < Kruskal-Wallis. This makes sense since the log10pct data was close to normal but did not have equal variances, matching most closely

to Welch's ANOVA.

3. Pseudocode You are working with a dataset of average daily tourism spending for five different types of visitor experiences in Connecticut: coastal, city/cultural, outdoor recreation, culinary, and family-oriented. Each observation represents a week's worth of spending data collected over the past three years. The dataset also includes information on total visitors, number of local events, and whether the data week included a holiday period. Your task is to determine whether average spending per visitor differs among experience types and to check whether assumptions of equal variances and normality of residuals are met.

- *Read in dataset*
- *Remove any rows with empty cells*
- *View the first few rows and get dim of dataset*
- *Make boxplots of average daily tourism (adt) according to visitor experience (ve)*
- *Make boxplots of $\log_{10}(\text{adt})$ according to visitor experience (ve)*
- *use aov() to compare mean of adt between ve and $\log_{10}(\text{adt})$ between ve and save to objects*
- *Do fits vs residuals and a qq plot to check for normality and equal variance in both aov objects*
- *Based on the plots, choose an appropriate model to fit (one-way ANOVA, Welch's ANOVA, or Kruskal-Wallis)*
- *Fit the selected model to determine whether average spending per visitor differs among experience types.*